Jiashu Han

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EDUCATION

Ph.D. in Physics, University of California, San Diego September 2026
Advisor: Daniel Green

B.A. in Physics and Astrophysics, University of California, Berkeley May 2019
with high distinction in general scholarship

TEACHING EXPERIENCE

Associate Instructor June 2026 — September 2026
University of California, San Diego

PHYS 1A: Mechanics (Summer Sessions I and II, 2026)

- Served as instructor of record for two offerings of introductory calculus-based mechanics intended for life science majors.
- Designed and delivered lectures covering kinematics, Newtonian mechanics, energy, momentum, and rotational motion.
- Developed instructional materials, assignments, and in-class assessments aligned with course learning objectives.
- Incorporated active-learning and formative-assessment strategies, including in-class conceptual questions and collaborative problem solving, to promote student engagement and conceptual understanding.
- Held regular office hours and provided individualized academic support to students with diverse levels of physics preparation, across two courses with total enrollment of more than 80 students.

Academic Associate / Lab Course Coordinator September 2021 — December 2025
University of California, San Diego

PHYS 1AL: Mechanics Laboratory (Winter 2022 — Fall 2025, instructors: A. Yagil, P. Tsai, J. Arlett, G. Bal, A. Frano)
PHYS 1CL: Waves, Optics, and Modern Physics Laboratory (Fall 2021, instructors: A. Yagil, P. Tsai)

- Supervised and trained more than 50 graduate teaching assistants and 60 undergraduate graders over 4 years.
- Coordinated more than 20 laboratory sections during each quarter and resolved instructional and administrative issues.
- Developed and revised laboratory manuals, assignments, and grading rubrics.
- Led regular instructional-team meetings to ensure consistency in lab instruction, grading, and assessment.

Teaching Assistant September 2019 — June 2026
University of California, San Diego

PHYS 1C: Waves, Optics, and Modern Physics (Winter 2026, instructor: P. Tsai)
PHYS 1AL: Mechanics Laboratory (Fall 2020)
PHYS 1BL: Electricity and Magnetism Laboratory (Winter 2020; Winter 2021)
PHYS 1CL: Waves, Optics, and Modern Physics Laboratory (Fall 2021)
PHYS 2A: Physics – Mechanics (Spring 2020, instructor: A. Burgasser)
PHYS 2B: Physics – Electricity and Magnetism (Spring 2021, instructor: O. Shpyrko)
PHYS 2C: Physics – Fluids, Waves, Thermodynamics, and Optics (Fall 2019, instructor: B. Shotwell)
PHYS 4B: Physics for Majors – Fluids, Waves, Statistical and Thermal Physics (Spring 2026, instructor: J. Hirsch)
PHYS 12: Energy and Environment (Winter 2026, instructor: V. Sharma)

- Independently led discussion sections and held office hours.
- Held substantial instructional responsibilities across multiple courses, including developing and grading homework assignments and exams.

Tutor / Academic Intern September 2017 — May 2018
University of California, Berkeley

COMPSCI 61B: Data Structures (Fall 2017)
PHYSICS 137A: Quantum Mechanics (Spring 2018)

TEACHING AWARDS AND RECOGNITION

Outstanding Graduate Teaching Scholar Award, UC San Diego SGTS Program November 2026
Distinguished Senior Graduate Teaching Award, Department of Physics, UC San Diego Spring 2026
Scholar in Student-Centered College Teaching, UC San Diego Teaching and Learning Commons Summer 2026

PEDAGOGICAL TRAINING AND PROFESSIONAL DEVELOPMENT

Practitioner in Student-Centered College Teaching Fall 2025
UC San Diego Teaching and Learning Commons

- 30 hours of pedagogical training in the Introduction to College Teaching (ICT) course on the use of evidence-based teaching practices and strategies to create an equitable learning experience for students.

SciCom Training February 2026
UC San Diego MRSEC & Fleet Science Center

- A 3-hour professional development workshop on science communication for informal audiences, emphasizing tools to create impactful, curiosity-driven science and engineering experiences at public engagement and outreach events.

Specialist in Student-Centered College Teaching Spring 2026
UC San Diego Teaching and Learning Commons

- 40 hours of advanced pedagogical training in the Advanced College Teaching (ACT) course focused on student-centered teaching practices and course design for instructors of record.
- Applied and reflected on evidence-based teaching practices through an independent instructor-of-record experience, with support from a community of scholarly instructors.

RESEARCH EXPERIENCE

Graduate Student Researcher May 2020 — September 2026
Department of Physics, University of California, San Diego

- Derived the likelihoods of primordial non-Gaussian signatures in the galaxy power spectrum, and demonstrated the practicality of using sample variance cancellation to significantly improve constraints on the signal amplitude in realistic experimental settings.
- Developed computational models and statistical inference tools to investigate the signatures of field interactions during the early universe using observational data from galaxy surveys.
- Examined previously unexplored parameter space of primordial non-Gaussianity and studied the sensitivity of current and future galaxy surveys to these novel signatures.

Summer Research Assistant June 2018 — August 2018
Center for Computational Astrophysics, Flatiron Institute

- Computed the cross-correlation between the *Planck* CMB gravitational lensing signal and the SDSS-IV/eBOSS quasars, and placed constraints on the SDSS-IV/eBOSS quasar bias while ruling out systematic effects.

Research Affiliate October 2016 — June 2018
Lawrence Berkeley National Laboratory

- Performed statistical inference on *Planck* CMB lensing and SDSS-III/BOSS galaxies and quasars.

PUBLICATIONS

- D. Green, **J. Han**, B. Wallisch, "Extending the Cosmological Collider: New Scaling Regimes and Constraints from BOSS," arXiv:2602.12232 [astro-ph.CO].
- D. Green, Y. Guo, **J. Han**, B. Wallisch, "Light Fields during Inflation from BOSS and Future Galaxy Surveys," *JCAP* **05** (2024) 090, arXiv:2311.04882 [astro-ph.CO].
- **J. Han**, S. Ferraro, E. Giusarma, S. Ho, "Probing Gravitational Lensing of the CMB with SDSS-IV Quasars," *Monthly Notices of the Royal Astronomical Society* **485** (2019) 2, 1720-1726, arXiv:1809.04196 [astro-ph.CO].

SKILLS

- Programming: Python (proficient), Java, C/C++, IDL, R
- Scientific computing: numerical modeling, simulation, optimization
- Data analysis: statistical modeling, parameter estimation, quantitative forecasting

RESEARCH INTERESTS

- Large-scale structure, cosmic microwave background
- Primordial non-Gaussianity, inflation